Thomas Pratt

Sydney, NSW | 0472 658 500 | thomaspratt2004@gmail.com
prattygf.github.io/portfolio | linkedin.com/in/thomas-gf-pratt

PROFILE

I'm a final-year Bachelor of Science student at the University of Sydney (Environmental Studies, Geology and Geophysics), graduating in 2026. I have experience with GIS and remote sensing, including spatial modelling in ArcGIS Pro and satellite image analysis in Google Earth Engine. I'm a team player and clear communicator, with experience leading group GIS work and training staff. I'm interested in GIS roles in environmental markets and natural resource management.

SKILLS

- **GIS:** ArcGIS Pro (ModelBuilder, multi-criteria analysis, raster and overlay analysis, cartography), QGIS
- **Remote Sensing:** Google Earth Engine, Sentinel-2 and Planet imagery, random forest, NDVI analysis
- **Programming:** Python (plate tectonics model simulation), JavaScript (Google Earth Engine)
- **Fieldwork:** geological mapping, field sketching and stratigraphic logging (Orange and Bombo, NSW)
- **Policy familiarity:** EPBC Act 1999, Native Title Act 1993, NSW SSD and SIA framework, ACCU Scheme
- **Communication:** explaining technical information in plain language to customers, council staff and in reports
- **Teamwork and leadership:** GIS lead in a five-person group project; train and supervise new staff
- **Working with people:** comfortable dealing with contractors, council staff and the general public

GIS AND REMOTE SENSING PROJECTS

Critical Minerals, Native Title and Social Impact Assessment, NSW **2026**
University of Sydney, group research project

- Contributed to GIS analysis of overlap between critical mineral developments/deposits and native title claims, alongside a review of NSW Social Impact Assessments for State Significant Developments.
- Conducted the literature review, synthesising 36 peer-reviewed sources on mining impacts under the Millennium Ecosystem Assessment framework.
- Found that 10 of 20 critical mineral mines in NSW lie on native title determination or registered claim land.

Jervis Bay Marine Park Zoning Plan **2026**
University of Sydney, five-person group project (GIS lead)

- Built the team's multi-criteria analysis model in ArcGIS Pro ModelBuilder to generate three alternative Sanctuary, Habitat and General Use zoning plans from a tourism stakeholder perspective.
- Combined six criteria layers (rocky reefs, seagrass and mangroves, commercial fishing grounds, wildlife buffers, boat ramp access and shipwrecks) with pairwise-weighted stakeholder priorities on a 10 m raster.
- Produced the final zoning maps and wrote the GIS methods section of the report.

Mangrove Dieback Analysis, Cabbage Tree Basin (Royal National Park) **2026**
University of Sydney, individual project

- Classified land cover (water, sand, saltmarsh, mangrove, trees) from Sentinel-2 imagery using a random forest model in Google Earth Engine, validated through accuracy assessment.
- Mapped about 7.5 ha of mangrove dieback (2019–2022) by NDVI differencing of Planet imagery, and identified major decline from late 2020 using a Sentinel-2 NDVI time series.

EDUCATION

University of Sydney **2024 - 2026**

- Bachelor of Science, majoring in Environmental Studies and minoring in Geology and Geophysics.

University of New South Wales **2023**

- Bachelor of Engineering (Honours), Software Engineering: completed first year, then transferred to USYD.

Knox Grammar School **2020 - 2022**

- Higher School Certificate, ATAR 95.05: ranked 1st in school in Earth and Environmental Science.

WORK EXPERIENCE

Stihl Shop Stanmore | *Senior Sales Associate* **August 2024 - Present**

- Advise council representatives and contractors on equipment for council maintenance and landscaping work.
- Recommend chainsaws, trimmers, blowers and safety gear matched to customer needs and provide post-purchase troubleshooting and maintenance advice.
- Open and close the store as a key holder and run the front of the store unsupervised.
- Train and supervise new staff on products, safety gear and the POS system.

The Athlete's Foot | *Retail Assistant* **March 2023 - April 2025**

- Produced sales reports and analysed product performance in MS Excel and Erply ERP.